

Lecture 11, MCQ + MCQ(1)

(L11)

1. a, ~~b~~, d.

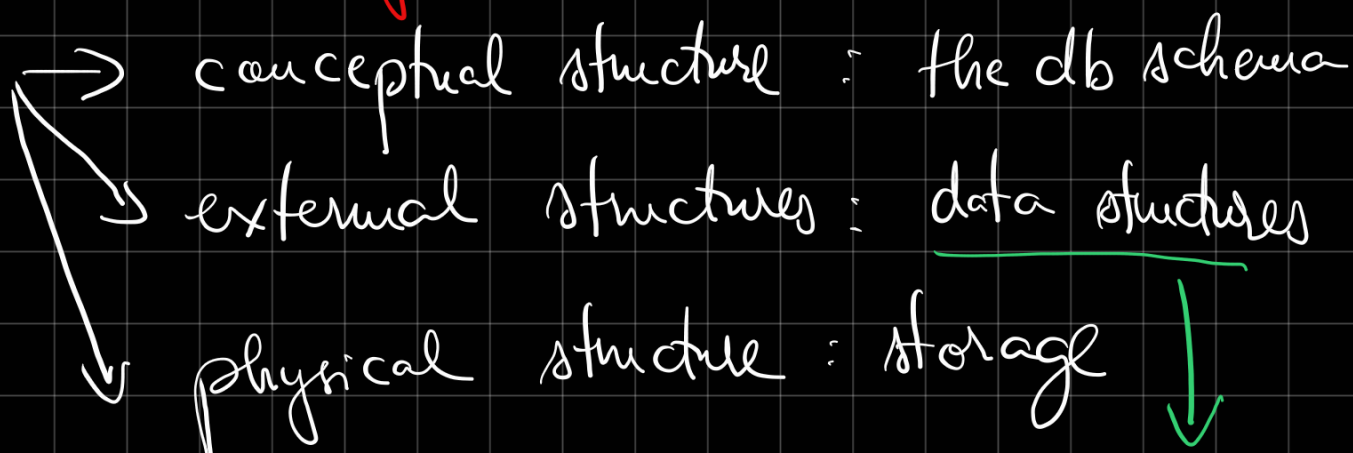
ORDER BY evaluated after the GROUP BY

2. b

Natural join of R_1 and R_2 contains all attributes from both relations, excluding duplicate columns.

3. c Multiple External Structures

The ANSI-SPARC architecture is a 3 level architecture for a database system



4. 1NF, ~~2NF~~ $\Rightarrow$ not 2NF $\Rightarrow$ C.

5. e.

6. a

7. c.

8.

$$Q_1: \left. \begin{array}{l} 1 \text{ null} \\ 1 \text{ null} \\ 1 \text{ null} \end{array} \right\} 3 + 2 \cdot 3 = 3 + 6 = 9$$

$$Q_2: 2 \cdot 3 = 6 \Rightarrow 9 - 6 = \boxed{3}$$

9. a

$$10. |Q_1| = 0, |Q_2| = 0 \Rightarrow \boxed{0}$$

11. b, c.

12.

	A	B
1	a ₁	b ₂
2	a ₁	b ₁
2	a ₂	b ₁

B	C
b ₂	c ₁
b ₁	c ₂
b ₁	c ₁

A	C
a ₁	c ₁
a ₁	c ₂
a ₂	c ₁

Same Data,
Different Views
for different
Users.

a c not in

$$4 - 3 = 1$$

2 2

AC

13. a, d

π .

①

T1
IDT1
C1

T2
IDT2
C2
IDT1 FOREIGN KEY

②

$$\pi_{\{A,B,C\}}((\pi_{\{A,B,ID\}}(\sigma_{M=70}(R))) \otimes_{ID=IDT1} (\pi_{\{C,IDT1\}}(\sigma_{N>5}(S))))$$

SELECT A,B,C

FROM (SELECT A, B, ID
FROM R
WHERE M=70) AS R1

INNER JOIN

(SELECT C, IDT1

FROM S

WHERE N>5) AS S1

ON R1.ID = S1.IDT1

OK (21.10 - 21.10.11)

3.

RID C2 C3
6 600 80000
2 202 1012

ANY $\Rightarrow$ MIN

4 400 40000
5 400 40000

4) $T = 4 * 6 = 24$

5) $\Rightarrow 200, 9 \rightarrow 2 \text{ results}$

RID	A	B	C	D	E	F
1	100	200	5	200	20	11
2	101	50	11	200	5	12
3	100	100	7	200	5	13
4	200	200	6	200	20	14
5	200	100	2	200	5	9
6	300	50	11	200	5	10

$\Rightarrow$ The whole R.

RID $\rightarrow$ A

6) $A \rightarrow D, B \rightarrow D, C \rightarrow D, E \rightarrow D, F \rightarrow D$

7) $(\bigcup_{R \in \text{join}} (R \times S)) - (\bigcup_{R \in \text{join}} (R \times S) \cap (T \times U))$

HCAQ :

1. e
2. d
3. c, d
4. a, c
5. b

6. c

7. b

8. b

9. a, b, c

10. a ?

11. NULL OR NULL $\Rightarrow$ NULL
c

The conceptual evaluation strategy
in a SELECT query is:

FROM, WHERE, GROUP BY,
HAVING, SELECT, ORDER BY.

HCAQ (2)

1. a, b, d

2. $2 \times 2 + 1 \times 1 + 2 \times 2 = 4 + 1 + 4 = \boxed{9} d$

3. e.

4. b

5. a, c

6. b

7. e

Natural join automatically joins tables based on the columns with the same name and data types.

Returns all columns from both tables with matching columns removed.

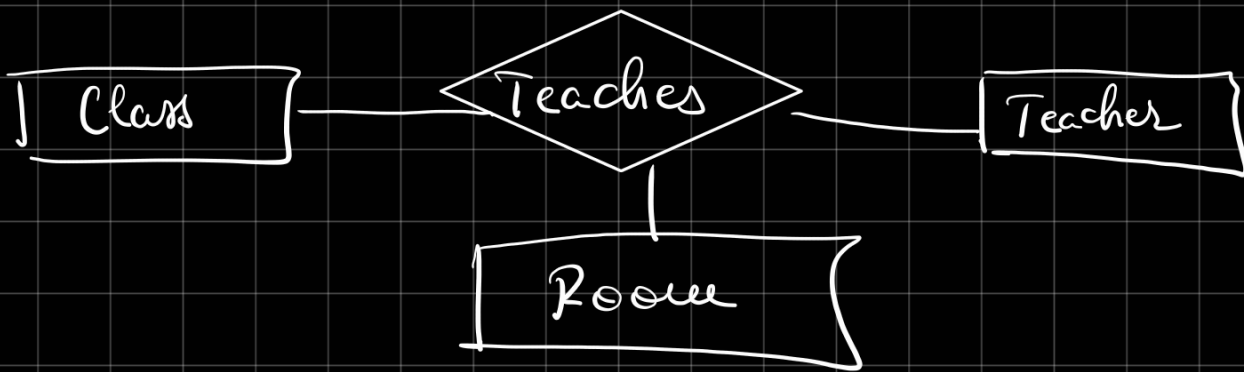
8. Not in 2 NF $\Rightarrow$ 1NF a)

9. b, c, d

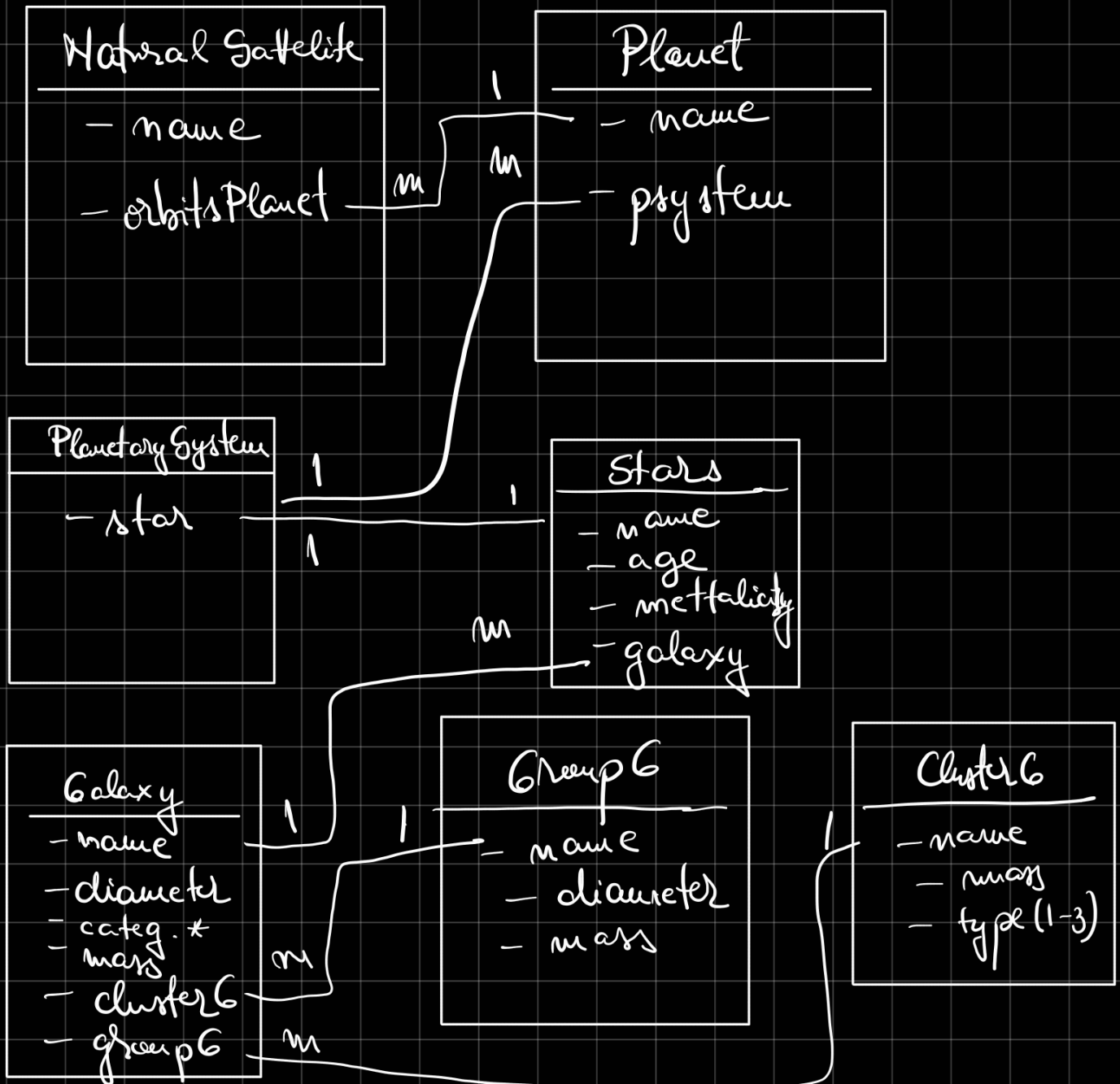
Exam Subjects 2020

I.1. The 3rd normal form represents a state of relation R where R is in 2nd normal form and all non-prime attributes depend on the key, the whole key and nothing but the key.

2. Ternary Relationship set = 3 entities



II.



b) CREATE TABLE Natural Satellite

(name VARCHAR(255) PRIMARY KEY,

orbitsPlanet VARCHAR(255) FOREIGN KEY
 REFERENCES Planets (name)
)

2. SELECT s.Name, s.Age

FROM Stars s

WHERE s.Galaxy = 'Milky Way' AND

(5 >= (SELECT COUNT(*)

FROM Planets p

WHERE p.PSystem = s.Name)

OR 1 >= (SELECT COUNT(*)

FROM NaturalSatellites n

INNER JOIN Planets pp

ON n.orbitsPlanet = pp.Name

WHERE pp.PSystem = s.Name)

)

b.

$\pi_{c.name, g.name} \left(\sigma_{c.type = 2} \left(\text{Galaxy} \otimes_{cluster6 = name} \text{ClusterG} \right) \right)$

III 1 a b d | 5 a, c

| Data Description

2. d

3. e

4. b

6. a, c, d

7. b

8. e

9. a

model describes:

- structure of data

- consistency constraints

- relationship with other data.

10. b, c, d

11. c, d?

Dense Indexes $\Rightarrow$ entries for every key in the dataset

Sparse Indexes $\Rightarrow$

- entries for portion of keys
- key - pointer pairs.
- reduced size of index.

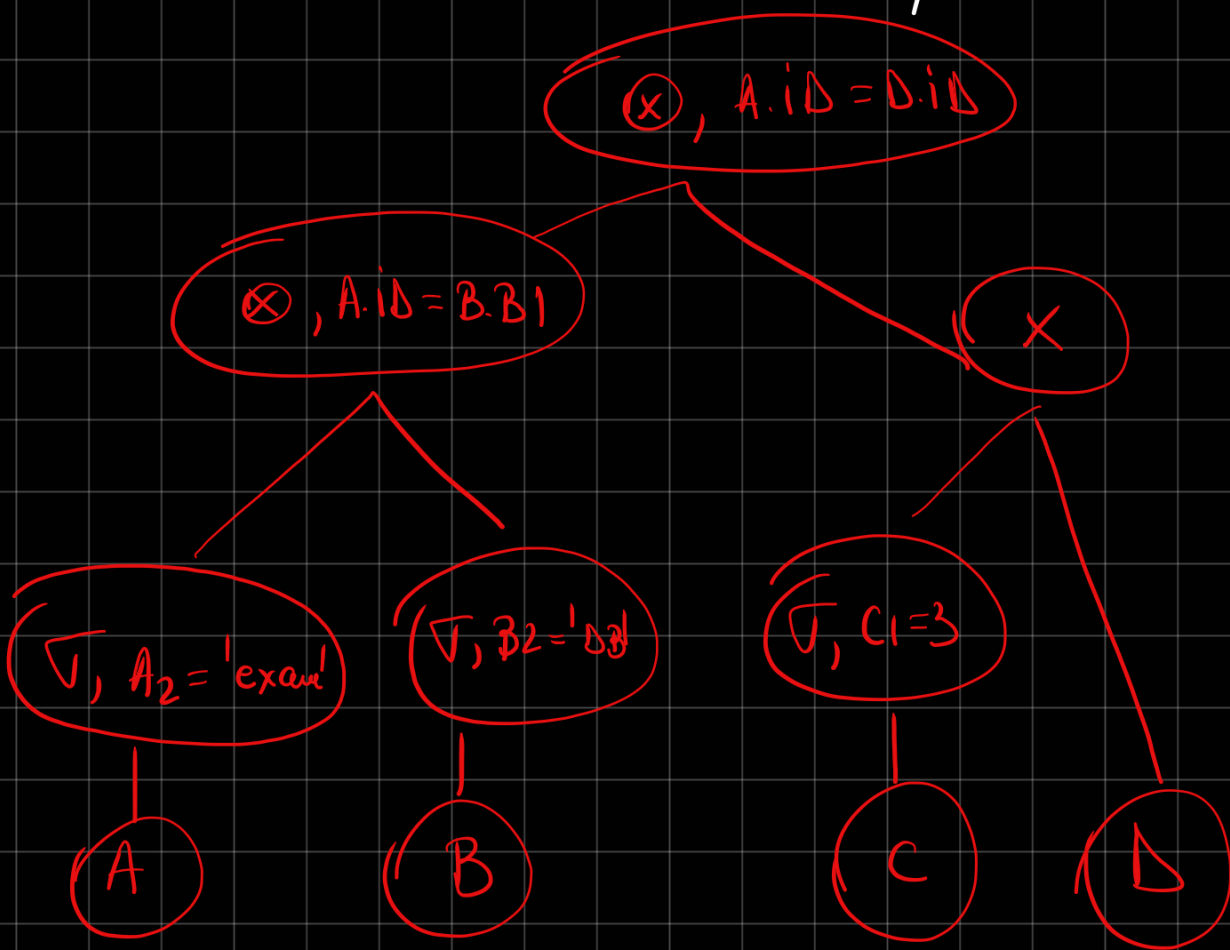
12. b, d

IV.

$$E = \left(\bigcup_{ID=B1 \text{ AND } A2='exam' \text{ AND } B2='DB'} (A \times B) \right)^* \left(\bigcup_{C1=3}^{(C \times D)} \right)$$

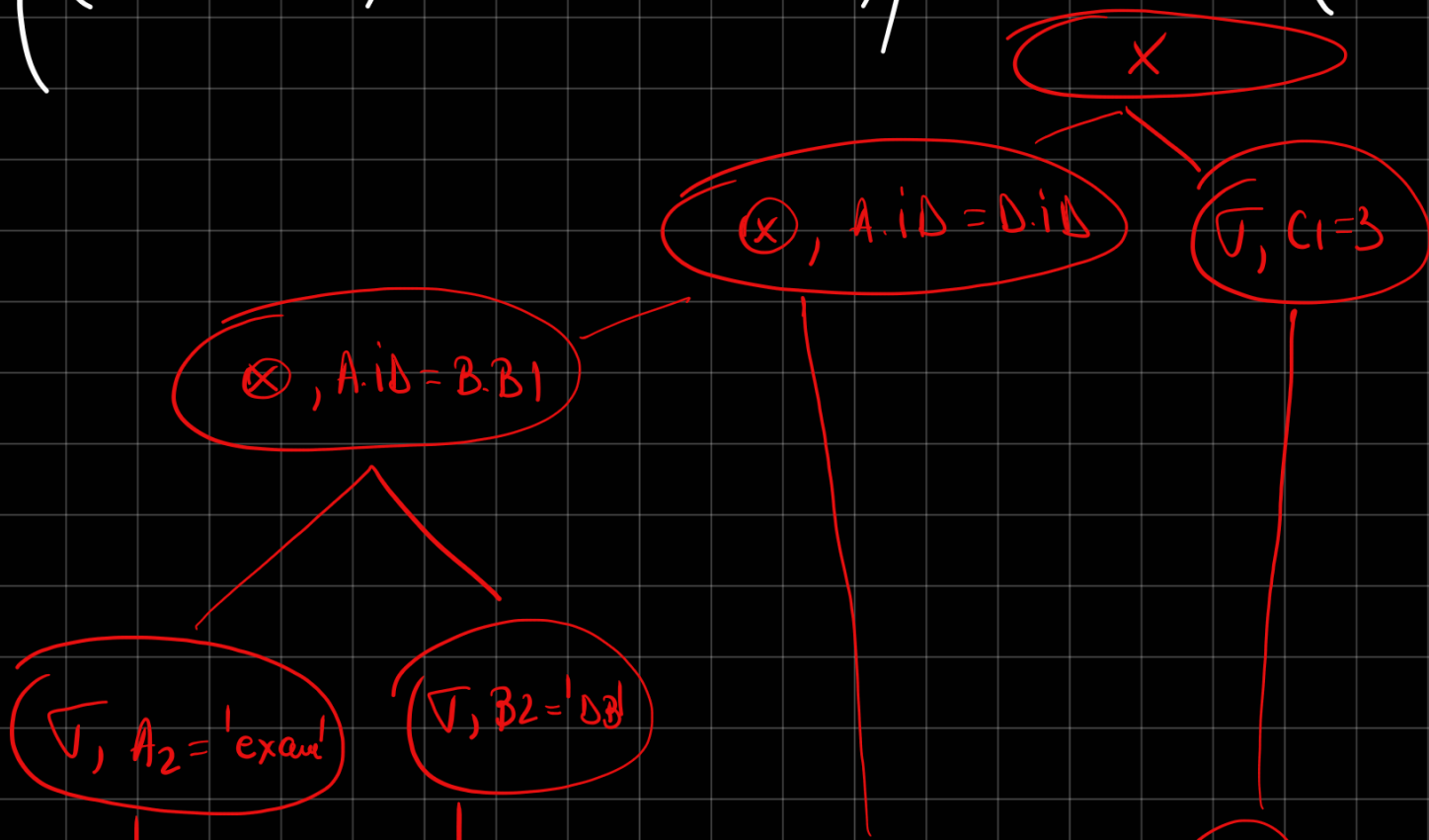
$$E = \left(\left(\bigcup_{A2='exam'} (A) \right) \otimes_{ID=B1} \left(\bigcup_{B2='DB'} (B) \right) \right)^* \left(\bigcup_{C1=3} (C) \times D \right)$$

$$E = \left(\left(\bigcup_{A2='exam'} (A) \right) \otimes_{ID=B1} \left(\bigcup_{B2='DB'} (B) \right) \right) \otimes_{A.IN=N.IN} \left(\bigcup_{C1=3} (C) \times D \right)$$



Q2

$$E = \left(\left(\left(\nabla_{A_2 = \text{'exam'}}(A) \right) \otimes_{iD = B1} \left(\nabla_{B2 = \text{'DB'}}(B) \right) \right) \otimes_{A.iD = D.iD} \left(X \left(\nabla_{C1 = 3}(C) \right) \right) \right)$$



A

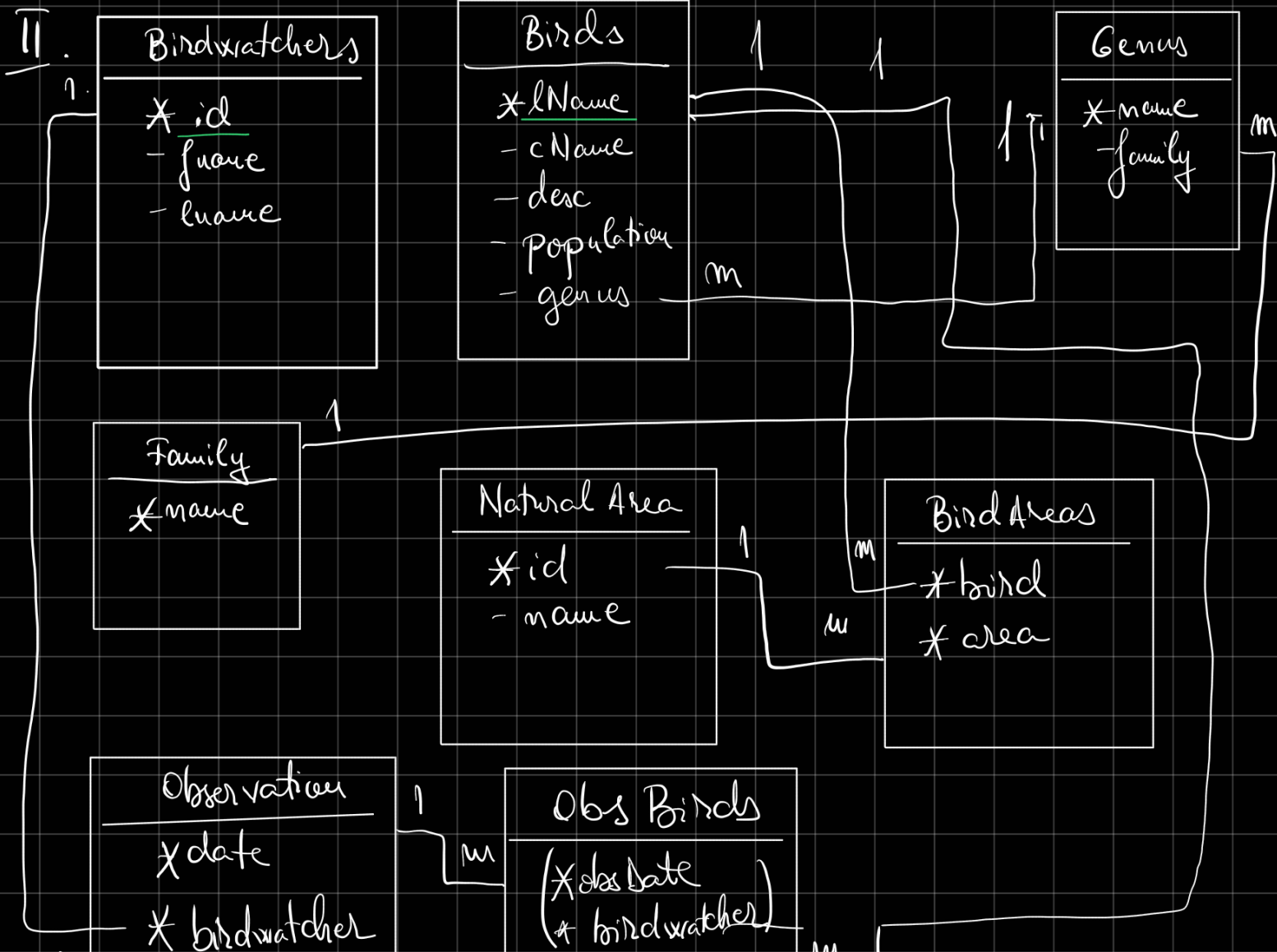
B

D

C

Model 2019

- I. 1. A key is a minimal subset of attributes that uniquely identifies a record.
2. A clustered index is an optimised/sorted index, where the order of the rows is structured for quick access. In which the logical order det. physical order.



x bind —

```
(obsDate DATETIME2,
```

bindwatcher int,

bind VARCHAR(255)

we INT NOT NUM

FOREIGN KEY (obsDate,

REFERENCES Observation (date, bird, index, etc.)

Dom: $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ $V \in V$ (ab Note bind, x, f, ch, bind)

)

(SELECT s.fname, l.fname, COUNT(*) as c1, b1.

FROM Observations or

INNER JOIN Bindv

GROUP BY o bindx/ratchers

GROUP BY o bindx/ratchers

HAVING COUNT(*) >= 10) s1

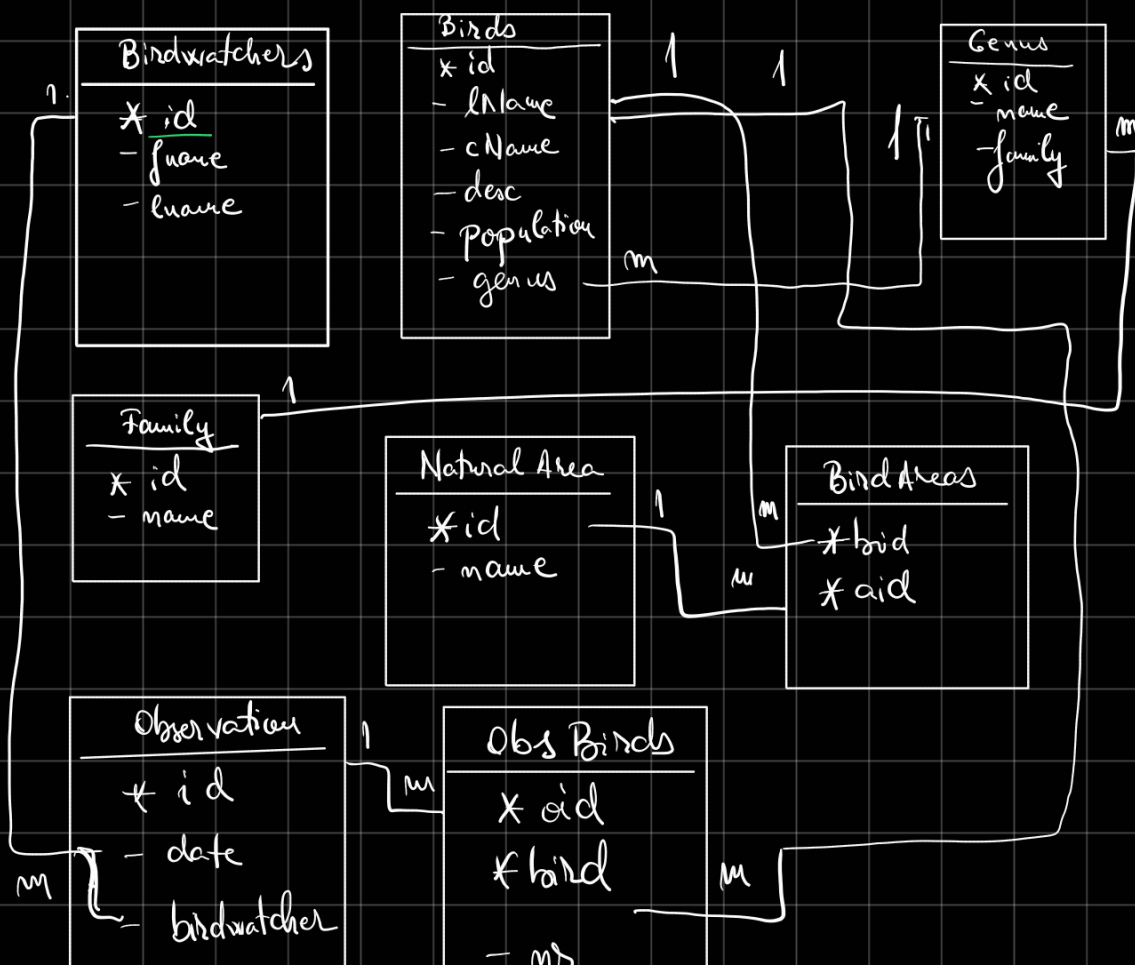
INNER JOIN

(SELECT COUNT(*) as C2, o1.date, o1.birdwatcher
FROM ObsBirds o1

INNER JOIN Birds b1 on o1.bird = b1.lname
WHERE b1.genus = 'Aquila') s2

on s1.date = s2.date AND s1.birdwatcher =
s2.birdwatcher.

1.a



2a)

```

SELECT t1.fname, t2.lname, t1.NrObs, t2.NrObs
FROM
(SELECT COUNT(*) as NrObs, b.fname, b.lname, b.id
FROM BirdWatchers b
INNER JOIN observations o ON b.id = o.birdwatcher
GROUP BY b.fname, b.lname, b.id
HAVING COUNT(*) >= 10)
t1
INNER JOIN
(SELECT oo.birdwatcher, count(*) as NrObs
FROM ObsBirds ob
INNER JOIN observations oo
ON ob.obid = oo.id
INNER JOIN Birds bb ON ob.bird = bb.id
INNER JOIN Genus g ON bb.genus = g.id
WHERE g.name = 'Aquila'
GROUP BY oo.birdwatcher) t2
ON t1.id = t2.birdwatcher

```

b.

$\pi_{cname, lname, pop}(\sigma_{pop < 1000 \text{ AND } id \in (\pi_{bid}(\text{BirdAreas}))})$ (Birds)

III. 1. a, d. 6. $20+21+22+23+24 \Rightarrow a$.

2. c

3. c

4. c

5. a.

7. a.

	A	B
2	A1	B1
1	A2	B2

B	C
B1	C1
B1	C2
B2	C1

A	C
A1	C1
A1	C2
A2	C1

$$\Rightarrow 3 - 3 = 0.$$

8. c, d

9. c

10. c

11. a

12. c, d

IV.

$$E = \sqrt{A.A1D = B.B1 \text{ AND } B.B1D = C.C2 \text{ AND } C.C3 = 5} \left(\pi_{\{A.A1D, A.A1, A.A2, B.B1D, B.B1, C.C2, C.C3\}} (A \times B \times C) \right)$$

$$E = \pi_{\{A1D, A1, A2\}}^{(A)} \otimes \pi_{B1D, B1}^{(B)} \otimes \pi_{C2, C3} \left(\sqrt{C3=5}^{(C)} \right)$$

$$(\otimes, B1D = C2)$$

$$(\otimes, A1D = B1)$$

